National University of Computer and Emerging Sciences



Lab Manual 03
Computer Organization
and Assembly Language

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Objectives:

- If-else, if-else-if, nested if
- Loops
- Arrays
- Unconditional/Conditional jumps.

Activities:

Activity 1 (Assembly Programming Tasks):

1. Write a program to find the absolute difference between two numbers. Store the result in a memory variable result. For example:
    a. if x > y, then result = x – y
    b. otherwise, result = y – x
2. Write a program to find the smallest number among three numbers. Store the smallest value in a memory variable smallest.
3. Write a program to classify a temperature value.
    Classification categories:
        i. < 0 → freezing (1)
        ii. 0–24 → cold (2)
        iii. 25–69 → moderate (3)
        iv. ≥ 70 → hot (4)
4. Write an assembly language program that compares two numbers stored in memory. If the first number is greater than the second, jump to a label that increments the first number by 1. Otherwise, jump to a label that decrements the first number by 1.
5. Write a program that continuously loops through a series of instructions until a specific condition is met. For this task, increment a counter until it reaches a certain value, and then use an unconditional jump to exit the loop.
6. Implement a bubble sort algorithm to sort an array of 5 numbers in ascending order using conditional and unconditional jumps.
    arr dw 5, 3, 8, 4, 1.
7. Implement a selection sort algorithm to sort an array of 6 numbers in descending order. Use jumps for the comparison and swapping logic.
8. Write a program that calculates the sum only for those numbers in an array that belong to the Fibonacci Series. For example:
    Array: 4, 5, 8, 13, 15, 21, 22, 34, 40, 55
    Sum: 5 + 8 + 13 + 21 + 34 + 55 = 136